Reg. No. | | | | | | | | | | | [13CE62/12CE62]

SIXTH SEMESTER B.Tech./B.E EXAMINATION, JUNE – 2016
RAILWAY AND AIRPORT ENGINEERING

Time: 3 Hours **Maximum Marks: 70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1. a. Discuss the factors governing the choice of gauge for a railway track. **06**

b. List the requirements of an ideal permanent way. **08**

2. a. List the functions of sleepers. Explain what is meant by sleeper density. List the values adopted on Indian Railways. **06**

b. Estimate the maximum permissible train load that a BG locomotive can haul with 4 pairs of driving wheels with axle load of 20 tonnes each along a straight level stretch at a speed of 80kmph. Calculate the reduction in speed on an up gradient of 1 in 150. Assess the further reduction in speed along a 3° curve on the up gradient. Assume co-efficient of friction as 0.16. **08**

3. a. Define creep of rails. What are the indications and effects of creep? **08**

b. On a BG track, a 6° curve diverges in an opposite direction from a 3° main curve. Calculate the allowable speed on the branch line. If the permissible speed on the main line in 65kmph. **06**

4. a. Draw a neat line diagram of a left hand turnout showing all the components. **06**

b. Explain the working procedure of automatic block system of controlling the movement of trains. **08**

5. a. List the factors to be considered for selecting a site for an airport. Explain what is meant by a preferential runway. **08**

b. Determine the best direction for orienting the runway from the wind data tabulated below. Estimate the percentage of time in a year during which the runway can be safely used for the following deviations of wind from the direction of landing and take off: **06**

i. 33.75°

ii. 56.25°

Wind Direction	Duration, percent	Wind Direction	Duration, percent	Wind Direction	Duration percent	Wind Direction	Duration percent
N	7.3	E	5.7	S	7.4	W	3.9
NNE	8.9	ESE	1.7	SSW	5.9	WNW	1.7
NE	14.6	SE	0.6	SW	10.5	NW	1.2
ENE	10.3	SSE	0.4	WSW	8.3	NNW	0.5

6.	**a.**	Describe the data to be collected for regional planning of a airports.	**06**
	b.	An airport is planned at an elevation of 380 m above MSL. The monthly mean of maximum and average daily temperatures for the hottest month at the site are 40°C and 28°C respectively. The effective gradient is 0.178 percent. Estimate the length of runway required at the proposed site if the basic runway length is 1900m.	**08**
7.	**a.**	Explain the various aircraft parking systems.	**08**
	b.	Illustrate the open parallel concept of airport layout indicating the various components.	**06**
8.	**a.**	Describe the Unit Terminal principle of terminal building planning and list the basic principles of passenger flow at airports.	**06**
	b.	Explain the flight procedure of ILS with the help of a schematic diagram.	**08**